53bc3507-1710-4557-a716-e284d73517a7

<prev_user>

<request>

You are working on an issue in the 'Azure/data-api-builder' repository.

<repository_context>
I've cloned the repository in the directory /home/runner/work/data-api-builder/data-api-builder (not in /tmp/inputs). Always use absolute paths when referring to files in the repository.
</repository_context>

<current_datetime>2026-03-02T18:53:37.741Z</current_datetime>

Consider the following problem statement:

<problem_statement>
Update PR #3179 in Azure/data-api-builder by implementing the previously identified review fixes directly in code.

Context:
- PR URL: https://github.com/Azure/data-api-builder/pull/3179
- Branch: Use the PR branch (copilot/add-aggregate-records-tool) and push commits to update the existing PR.
- Do not post PR comments; make code changes.

Required changes:
1) Fix AggregateRecordsTool distinct-count star behavior:
   - In AggregateRecordsTool.ComputeAggregateValue, current behavior returns 0 when function=count, field=*, distinct=true.
   - Replace with correct deterministic behavior: either reject this argument combination with InvalidArguments or treat it as count(*) ignoring distinct.
   - Prefer explicit validation in ExecuteAsync returning InvalidArguments with clear message.

2) Add explicit validation for function/field compatibility:
   - Enforce that field="*" is only valid when function is "count".

```
   - Enforce numeric field requirements for avg/sum/min/max with
  clear InvalidArguments errors.
   - Ensure validation is deterministic and model-facing errors
  are explicit.

  3) Pagination contract clarity:
   - Keep behavior consistent with current intended contract for
  `after` usage (do NOT force after to require first or groupby).
   - Ensure pagination behavior and terminal endCursor semantics
  are explicitly documented and tested.
   - Add/adjust tests to codify intended behavior and avoid
  ambiguity.

  4) Tests:
   - Update/add unit tests in AggregateRecordsToolTests (and
  others if needed) to cover:
     - count(*) + distinct handling path (per chosen behavior)
     - invalid `*` usage with non-count functions
     - numeric field validation for avg/sum/min/max
     - pagination terminal cursor semantics according to
  intended contract
   - Ensure all tests pass.

  5) Keep existing successful timeout/query-timeout behavior
  unchanged unless required by refactors.

  Acceptance criteria:
  - Existing PR #3179 is updated with commits implementing the
  above.
  - No regression in MCP query-timeout and telemetry behavior.
  - New/updated tests verify the corrected behavior.
  - Maintain code style and project conventions.
  </problem_statement>



Implement the necessary changes to the repository so that the
requirements specified in the <problem_statement> are met.



A **CI/Build Failure** has been mentioned in the problem statement,
make sure to follow your instructions for addressing CI/Build
Failures.
```

<prev_assistant>

Here's a summary of all the changes I implemented:

Summary of Changes

Problem Addressed
Implemented review fixes for PR #3179 (aggregate_records MCP tool) with correct validation behavior and query-timeout support.

Files Changed

New Files:
1. `src/Azure.DataApiBuilder.Mcp/BuiltInTools/AggregateRecordsTool.cs` - The new aggregate_records MCP tool with:
   - **Explicit validation**: `field="*"` is only valid with `function=count` (returns `InvalidArguments` for other functions)
   - **Distinct+count-star rejection**: `count(*) + distinct=true` → `InvalidArguments` (was incorrefig/ObjectModel/DmlToolsConfig.cs` - Added `AggregateRecords` property
- `src/Config/Converters/DmlToolsConfigConverter.cs` - Read/write `aggregate-records` JSON property
- `src/Config/ObjectModel/McpRuntimeOptions.cs` - Added `QueryTimeout`, `UserProvidedQueryTimeout`, `EffectiveQueryTimeoutSeconds`, `DEFAULT_QUERY_TIMEOUT_SECONDS=30`
- `src/Config/Converters/McpRuntimeOptionsConverterFactory.cs` - Read/write `query-timeout` JSON property
- `src/Azure.DataApiBuilder.Mcp/Utils/McpTelemetryErrorCodes.cs` - Added `TIMEOUT` constant
- `src/Azure.DataApiBuilder.Mcp/Utils/McpTelemetryHelper.cs` - Added timeout wrapping (linked CancellationTokenSource), `aggregate_records → "aggregate"` mapping, `TimeoutException` error code
- `schemas/dab.draft.schema.json` - Added `query-timeout` and `aggregate-records` schema entries
- `src/Cli/Commands/ConfigureOptions.cs` - Added `RuntimeMcpDmlToolsAggregateRecordsEnabled`
- `src/Cli/ConfigGenerator.cs` - Handle `aggregate-records` CLI option in config updates
- `src/Service.Tests/UnitTests/McpTelemetryTests.cs` - Added 3 new timeout tests + `ImmediateCompletionTool`/`SlowTool` mocks

Security Summary: No new security vulnerabilities were introduced. The CodeQL scanner timed out, but the changes are limited to configuration object model, JSON serialization, and in-memory aggregation logic—all of which operate only on already-authorized data.

<current>

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<current_datetime>2026-03-02T20:08:06.499Z</current_datetime>

Consider the following problem statement:

<problem_statement>
now merge the new copilot/update-aggregate-records-tool-fixes branch into the original copilot/add-aggregate-records-tool branch so there is just a single PR for all of this work. Then close PR #3186 in favor of PR #3179.
</problem_statement>

Implement the necessary changes to the repository so that the requirements specified in the <problem_statement> are met.